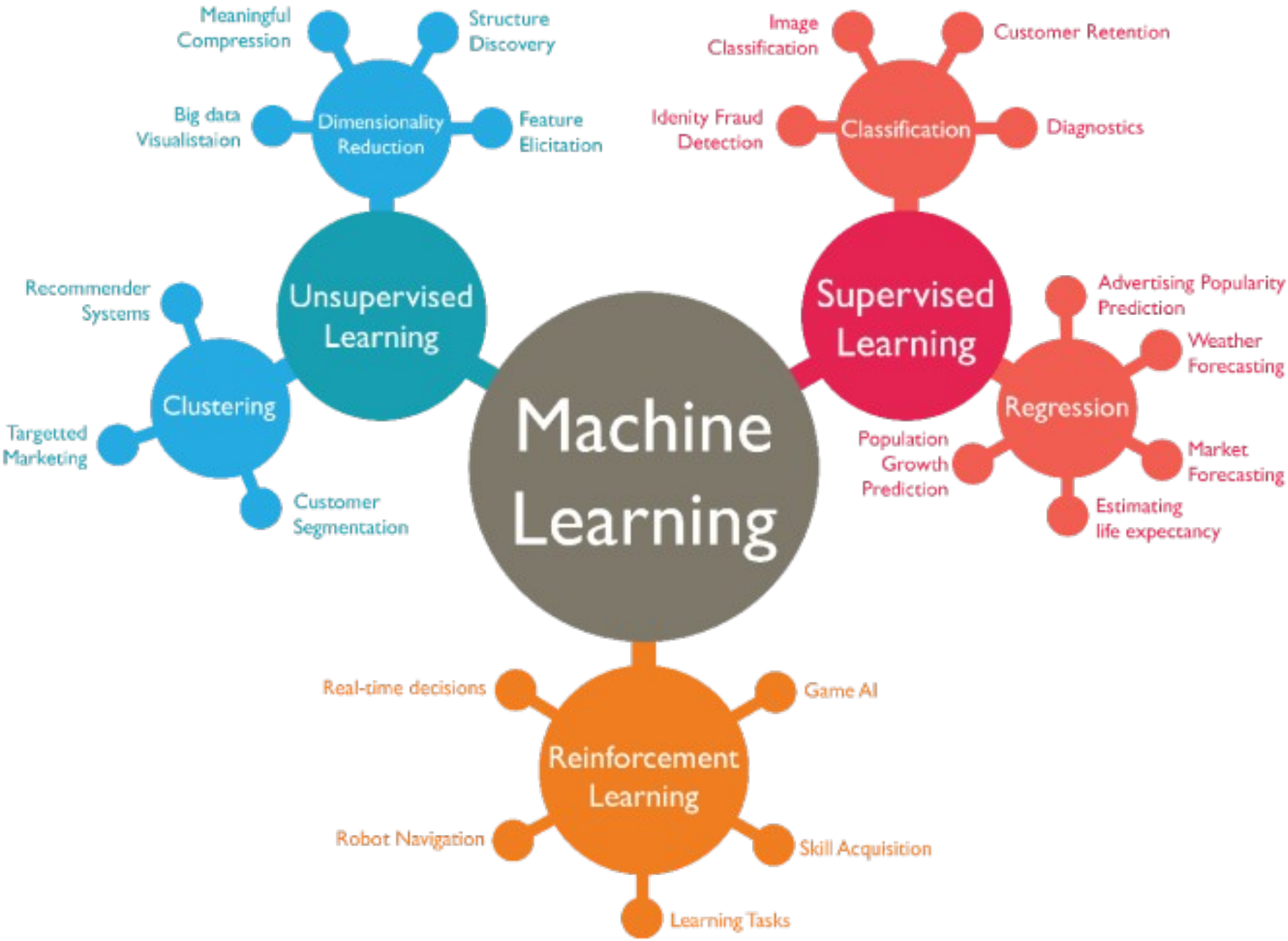
How to train the model?

What are the types of machine learning algorithms?


Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Reinforcement Learning
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

Supervised learning

Supervised learning

Input: the training set
questions and answers
data + labels

Classification
Regression

KAREN ZACK/@TEENYBISCUIT

kaggle.com/c/dog-breed-identification

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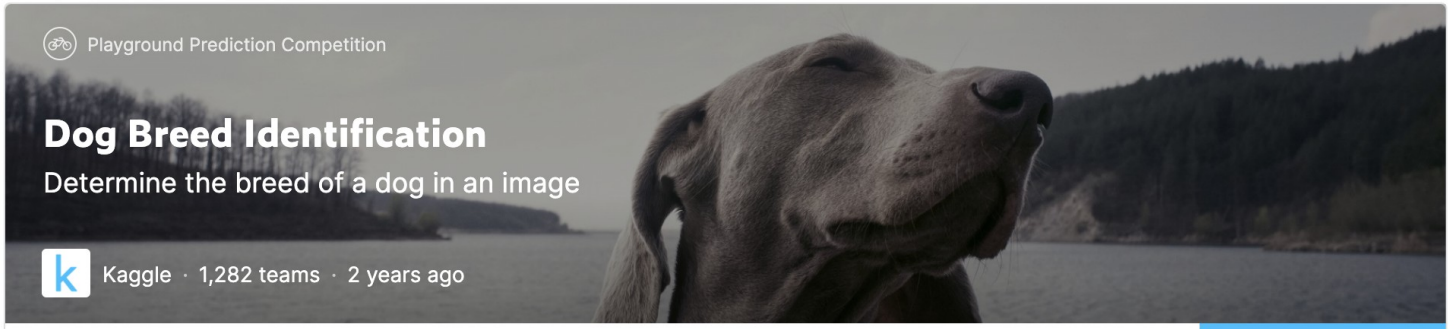


Overview Data Notebooks Discussion Leaderboard Rules

Join Competition

Overview

Description

Evaluation

Who's a good dog? Who likes ear scratches? Well, it seems those fancy deep neural networks don't have *all* the answers. However, maybe they can answer that ubiquitous question we all ask when meeting a four-legged stranger: what kind of good pup is that?

In this playground competition, you are provided a strictly canine subset of ImageNet in order to practice fine-grained image categorization. How well you can tell your Norfolk Terriers from your Norwich Terriers? With 120 breeds of dogs and a limited number training images per class, you might find the problem more, err, ruff than you anticipated.

labels.csv (470.76 KB)

Detail Compact Column 2 of 2 columns

About this file

This file does not have a description yet.

id	breed
10222 unique values	scottish_deerhound 1% maltese_dog 1% Other (9979) 98%
000bec180eb18c7604dcecc8fe0dba07	boston_bull
001513dfcb2ffafc82ccf4d8bbaba97	dingo
001cdf01b096e06d78e9e5112d419397	pekinese
00214f311d5d2247d5dfe4fe24b2303d	bluetick
0021f9ceb3235effd7fcde7f7538ed62	golden_retriever
002211c81b498ef88e1b40b9abf84e1d	bedlington_terrier

Data Explorer

715.67 MB

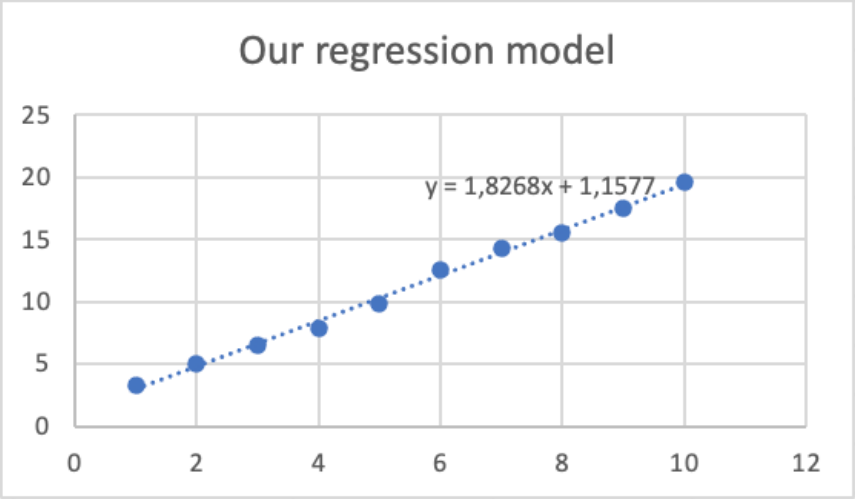
- test
- train
- labels.csv
- sample_submission.csv

Summary

- 20.6k files
- 123 columns

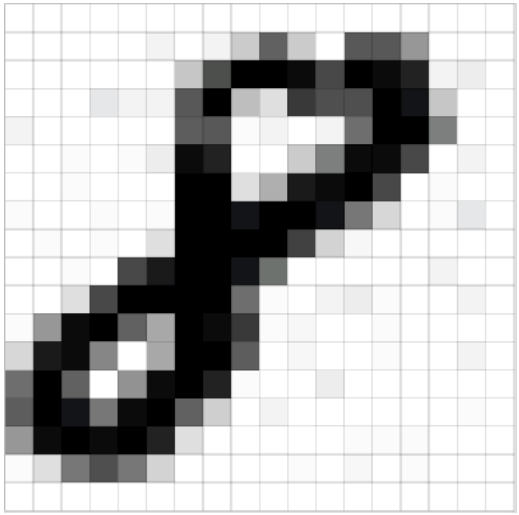
Download Output

x	y
1	3,22404622
2	5,03699641
3	6,55245888
4	7,88049804
5	9,87717329
6	12,5849065
7	14,280954
8	15,4925002
9	17,508476
10	19,6150603



Input: matrices

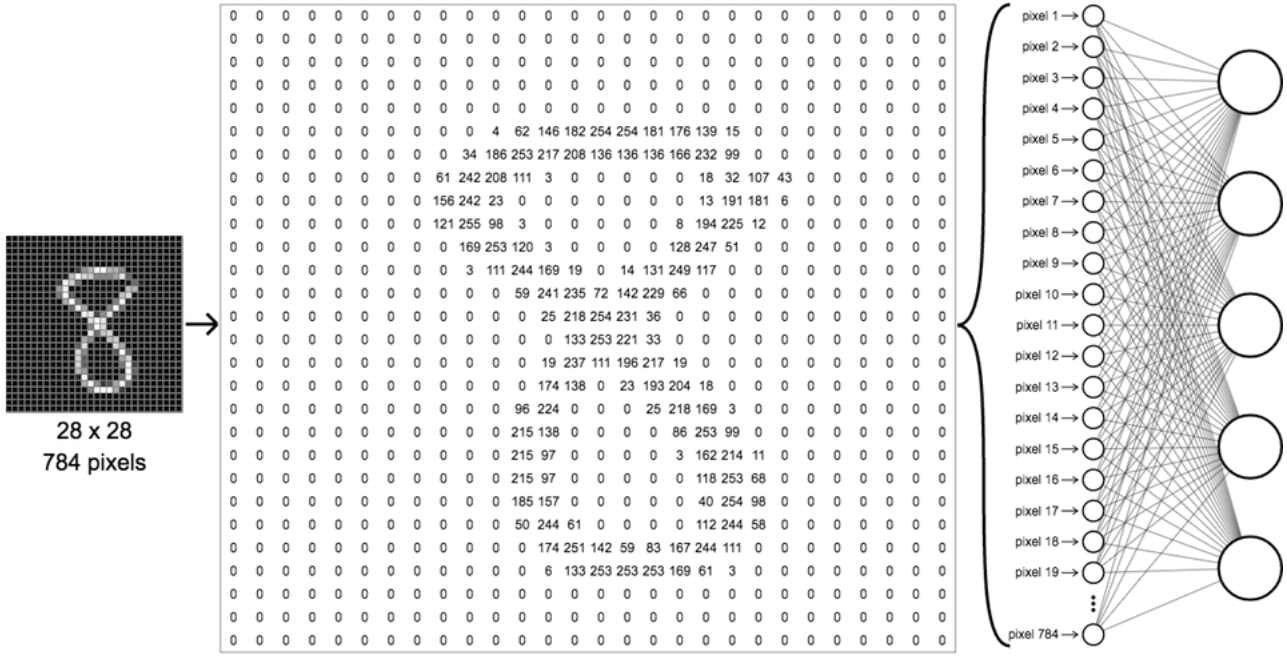
Each image can be saved as a matrix, with each number = pixel characteristics (0-255)

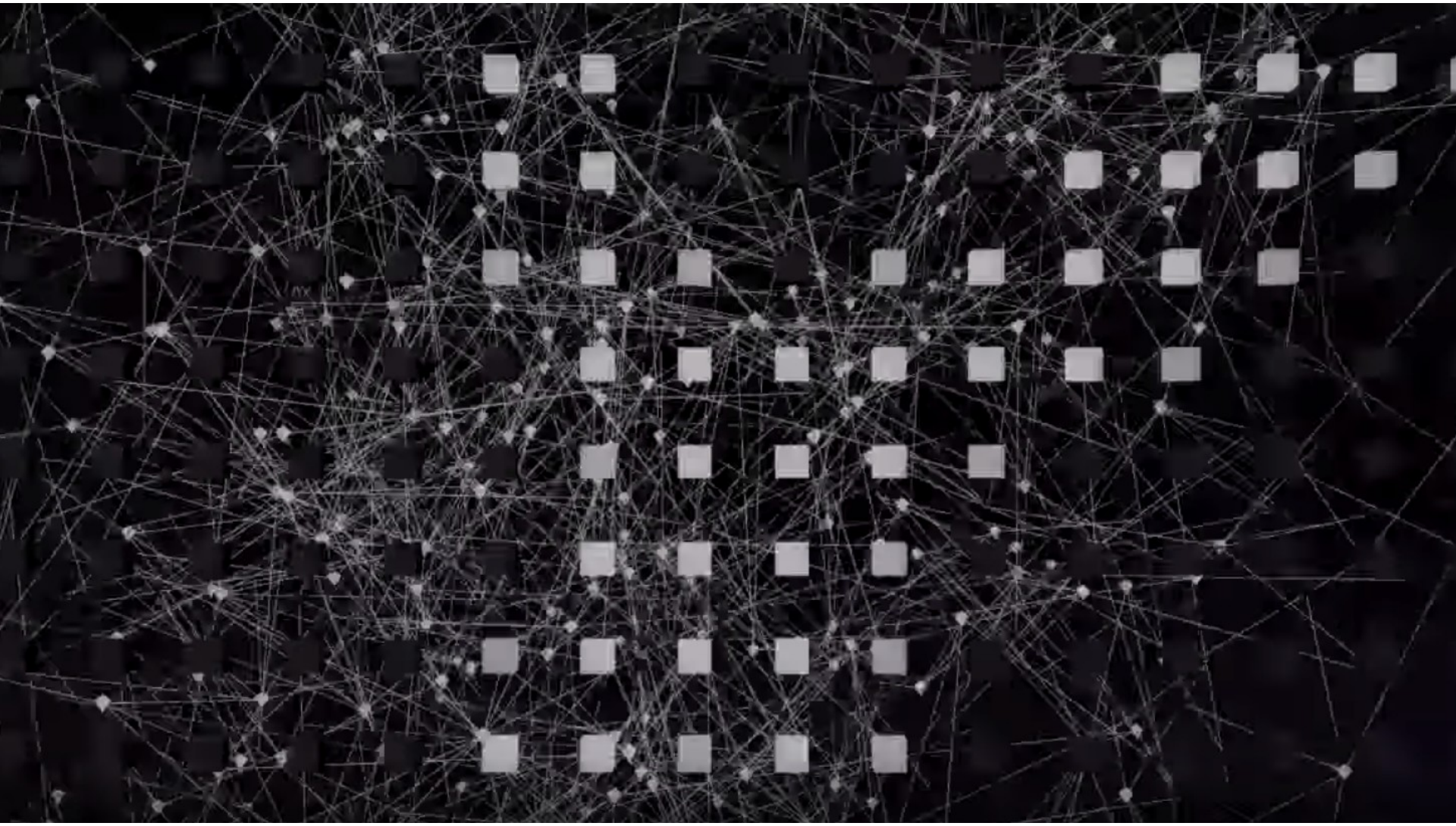


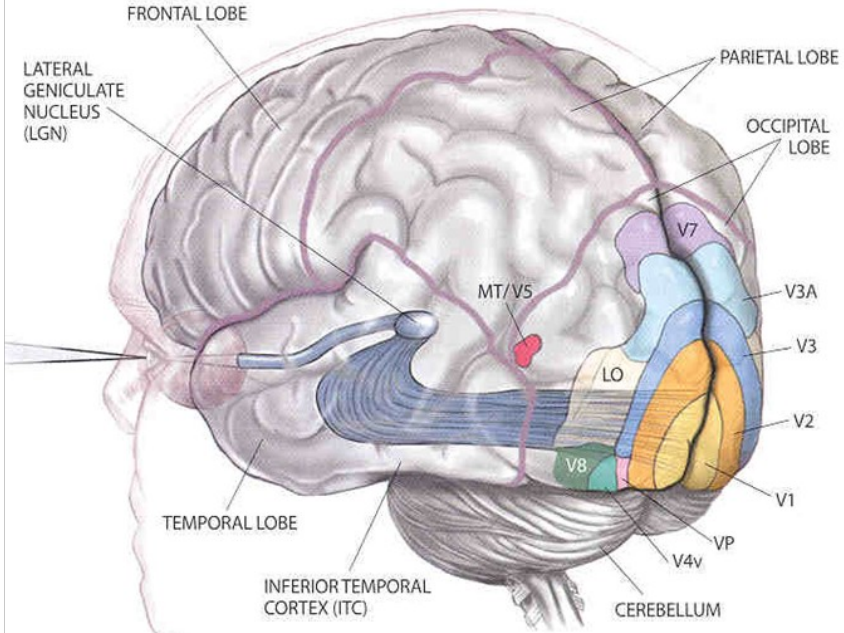
https://ujjwalkarn.me/2016/08/11/intuitive-explanation-convnets/

Output: categories

Exactly: the probability that an object belongs to this category



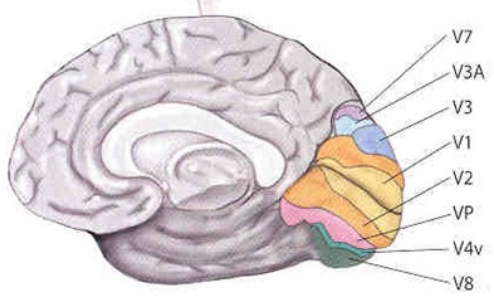


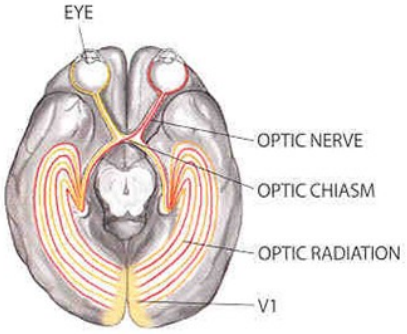


KEY TO FUNCTION

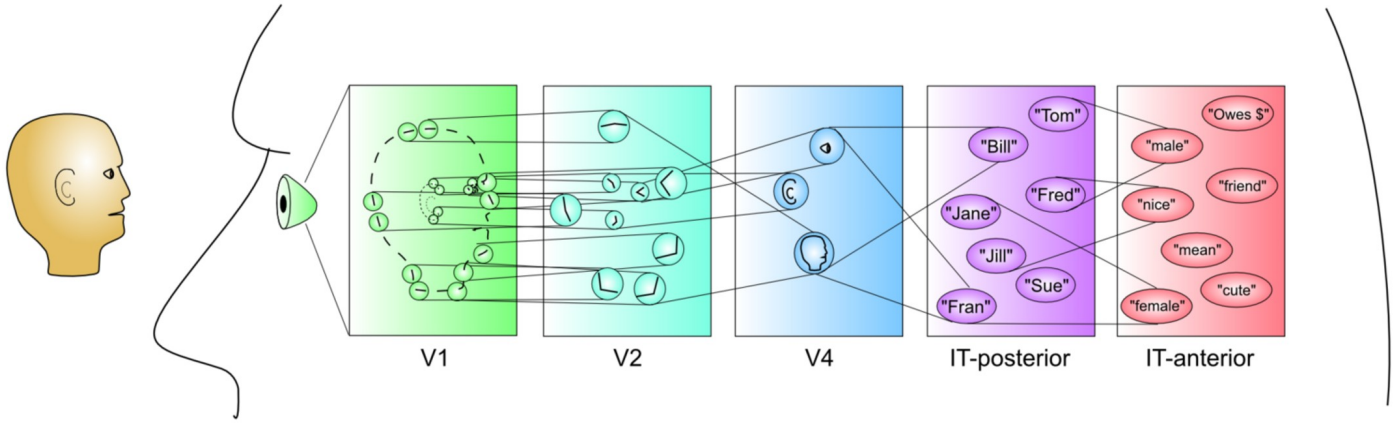
- **V1:** Primary visual cortex; receives all visual input. Begins processing of color, motion and shape. Cells in this area have the smallest receptive fields.
- **V2,** **V3** and **VP:** Continue processing; cells of each area have progressively larger receptive fields.
- **V3A:** Biased for perceiving motion.
- **V4v:** Function unknown.
- **MT/V5:** Detects motion.
- **V7:** Function unknown.
- **V8:** Processes color vision.
- **LO:** Plays a role in recognizing large-scale objects.

Note: A V6 region has been identified only in monkeys.





TERESE WINSLOW WITH ASSISTANCE FROM NOUCHINE HADJIKHANI AND ROGER TOOTELL *Harvard Medical School*



https://grey.colorado.edu/CompCogNeuro/index.php/CCNBook/Perception